YOLO Object Recognition and Tracking (PTZ)

1. Experiment Objective

Click any YOLO-detected object with the mouse, and the car body rotates to keep tracking it.

2. Model and Configuration Check

The YOLO ONNX follow node reads the configuration file by default:

```
~/microros_ws/src/yolo_onnx_follow_pkg/config/yolo_onnx_follow_config.yaml
```

Focus on the following parameters:

```yaml
YOLO_ONNX_FOLLOW:
  model_path: "/home/yahboom/MODELS/yolo/yolo11n.onnx"
  image_topic: "/camera/image_raw"
  input_width: 640
  input_height: 640
  conf_threshold: 0.2
  nms_threshold: 0.45
  max_detections: 3
```

If the model path on your machine differs (you can replace it with your own trained model), modify `model_path`, or override it at launch with `model_path:=`:

```
ros2 launch yolo_onnx_follow_pkg yolo_onnx_follow.launch.py
model_path:=/home/yahboom/MODELS/yolo/yolo11n.onnx
```

Replacing the Model with a Custom One

In general, if you only switch to a standard YOLOv8/YOLOv11 ONNX model, you do not need to change the Python source code; you only need to edit the configuration file:

- `model_path`: change it to your own `.onnx` model path.
- `class_names_path`: change it to your own class names file path.
- `input_width`, `input_height`: change them to the input size used when exporting the ONNX model.
- `target_class` or `target_class_id`: set this if you want to auto-follow a certain class of target.
- `conf_threshold`: lower it if your self-trained model misses detections; raise it if there are too many false positives.

There are mainly two cases that require changing the source code.

First: if you want to hard-code the class names directly in the program instead of using `class_names_path`, modify the `COCO_CLASS_NAMES` at the top of `yolo_onnx_follow_pkg/yolo_onnx_follow_node.py`:

```
COCO_CLASS_NAMES = [
    "person",
    "cup",
    "bottle",
]
```

Using `class_names_path` is recommended instead, so you do not need to change the source code when switching models.

3. Experiment Steps

1. Hardware Preparation

Hardware	Requirement
PTZ version	✗
No-PTZ version	☑
Required peripherals	ESP32 camera (fixed on a bracket)

Change the car model in the `.bashrc` file to the corresponding model:

```
gedit .bashrc
```

```
export CAR_TYPE=ptz       #ptz | no_ptz | basic_car   三种变量
#echo "LOCAL_IP: $(hostname -I | awk '{print $1}')"
echo "========== Current IP =========="
#ip -o -4 addr show scope global | awk '{split($4,a,"/"); printf "%-10s %s\n", $2":", a[1]}'
ip -o -4 addr show scope global | awk '{
    split($4,a,"/");
```

After editing, save and exit, then run the following command to apply the changes:

```
source ~/.bashrc
```

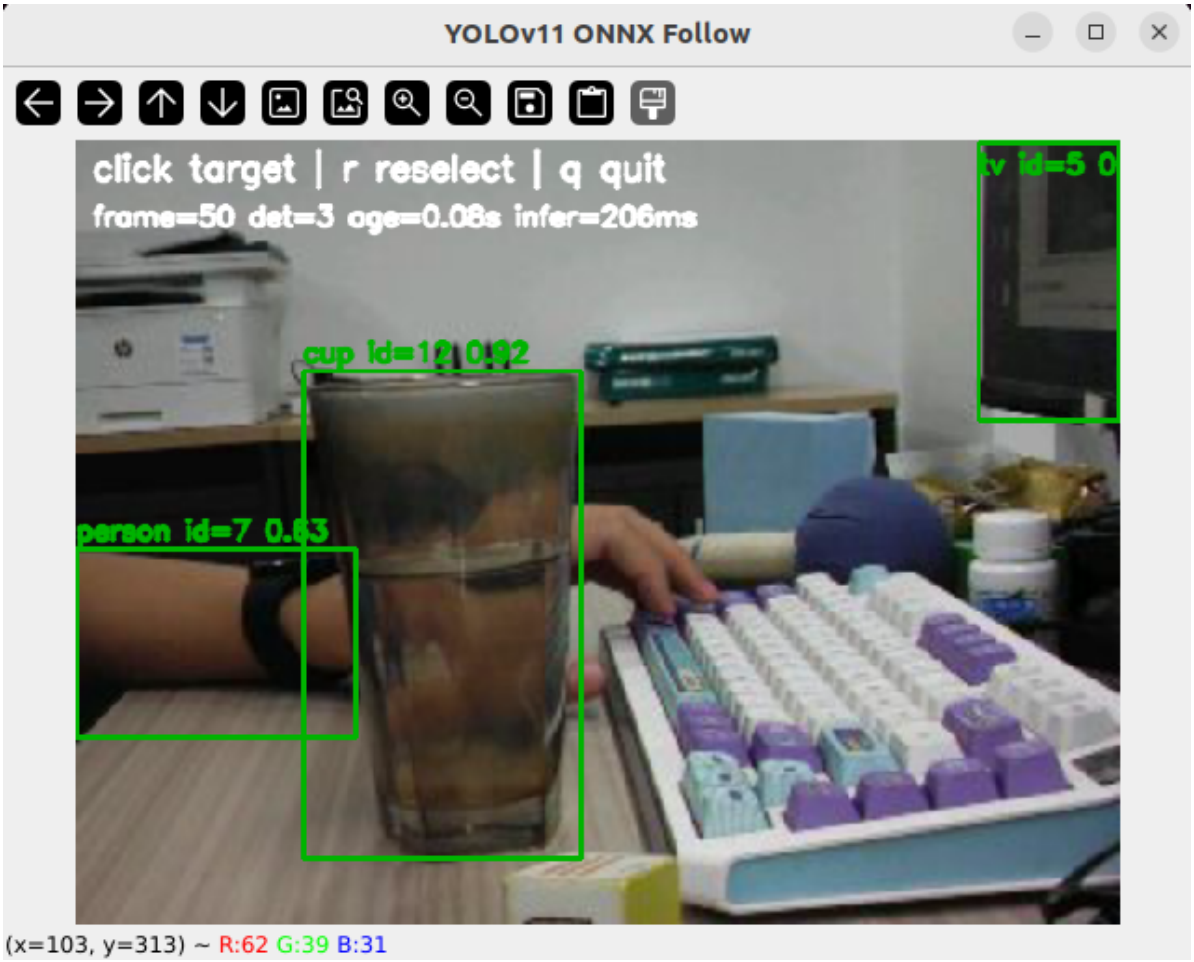
2. Start Agent + Camera + Joint Model (Terminal 1)

```
ros2 launch microros_v2_bringup car_base.launch.py
```

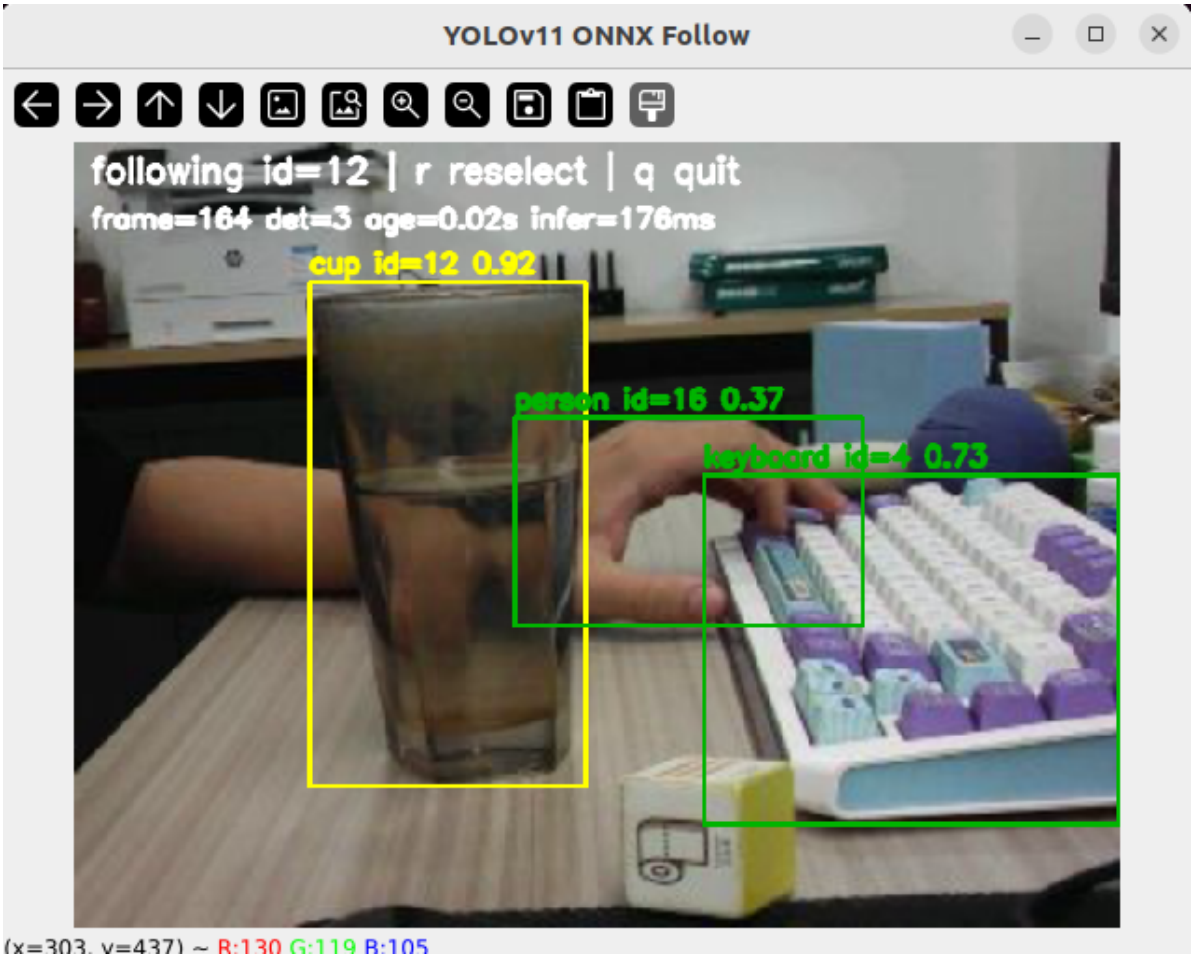
```
yahboom@yahboom:~$ ros2 launch microros_v2_bringup car_base.launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-07-28-15-04-46-269642-yahboom-579370
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [micro_ros_agent-1]: process started with pid [579411]
[INFO] [camera_driver-2]: process started with pid [579413]
[INFO] [robot_state_publisher-3]: process started with pid [579415]
[INFO] [joint_state_publisher-4]: process started with pid [579417]
[INFO] [ekf_node-5]: process started with pid [579419]
[micro_ros_agent-1] [1785222287.404236] info     | UDPv4AgentLinux.cpp | init                   | running...             | port: 8090
[robot_state_publisher-3] [INFO] [1785222288.847816851] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-3] [INFO] [1785222288.848013042] [robot_state_publisher]: got segment base_link
[robot_state_publisher-3] [INFO] [1785222288.848028942] [robot_state_publisher]: got segment bwheel_Link
[robot_state_publisher-3] [INFO] [1785222288.848037122] [robot_state_publisher]: got segment cam1_Link
[robot_state_publisher-3] [INFO] [1785222288.848043874] [robot_state_publisher]: got segment cam2_Link
[robot_state_publisher-3] [INFO] [1785222288.848050538] [robot_state_publisher]: got segment cam3_Link
[robot_state_publisher-3] [INFO] [1785222288.848057264] [robot_state_publisher]: got segment imu_frame
[robot_state_publisher-3] [INFO] [1785222288.848063747] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-3] [INFO] [1785222288.848070240] [robot_state_publisher]: got segment lfwheel_Link
[robot_state_publisher-3] [INFO] [1785222288.848076888] [robot_state_publisher]: got segment rfwheel_Link
[joint_state_publisher-4] [INFO] [1785222289.372349515] [joint_state_publisher]: Waiting for robot_description to be published on the robot_description
[camera_driver-2] [INFO] [1785222289.716161193] [esp32_ip_camera_publisher]: camera node started, camera_url: http://192.168.12.37:81/stream
[camera_driver-2] [WARN] [1785222292.705741414] [esp32_ip_camera_publisher]: Camera not opened, trying reconnect... (next retry in 1.0s)
[camera_driver-2] [INFO] [1785222294.072701944] [esp32_ip_camera_publisher]: IP camera stream opened successfully
```

3. Start Object Recognition and Tracking (Terminal 2)

```
ros2 launch yolo_onnx_follow_pkg yolo_onnx_follow.launch.py
```



Click the object you want to follow in the detected results with the mouse:



Operation Instructions

After the program starts, an OpenCV window pops up showing the detection boxes, class names, tracking IDs, and confidence scores.

- Left-click a target box: select and start following that target.
- `r`: clear the current target, stop the chassis turning, and re-select a target.
- `q` or `Esc`: exit the program; on exit it publishes a stop velocity and returns the pan-tilt to its initial angle.

Normal operation:

- Green detection boxes appear in the window.
- The target box turns yellow after being clicked.
- When the target deviates from the frame center, the pan-tilt turns first to correct.
- When the pan-tilt horizontal angle deviates a lot, the chassis publishes `angular.z` to turn in place.
- After the target-lost time exceeds the threshold, the chassis stops turning; if a target of the same class reappears nearby within a short time, the program tries to re-acquire it.

Now slowly move the recognized object, and the pan-tilt and car chassis will track it.

4. Source Code Analysis

4.1 File Structure

```
yolo_onnx_follow_pkg/
├── launch/
│   └── yolo_onnx_follow.launch.py
├── config/
│   └── yolo_onnx_follow_config.yaml
├── yolo_onnx_follow_pkg/
│   ├── yolo_onnx_follow_node.py
│   └── ptz_controller.py
├── setup.py
└── package.xml
```

4.2 Configuration File

`config/yolo_onnx_follow_config.yaml` is divided into three main sections:

- `SYSTEM`: configures the chassis velocity topic, `/cmd_vel` by default.
- `PTZ`: configures the pan-tilt angle topic, servo limits, initial angle, publish count, and whether to recenter after stopping.
- `YOLO_ONNX_FOLLOW`: configures the model path, image topic, detection thresholds, window display, target selection, tracking re-acquisition, and control parameters.

Commonly tuned parameters:

Parameter	Description
`conf_threshold`	Detection confidence threshold; lower values detect more easily but also cause more false positives
`max_detections`	Maximum number of high-confidence detection boxes kept per frame
`max_inference_rate_hz`	Maximum inference rate; lower it when performance is insufficient
`follow_rate_hz`	Follow control rate
`pixel_tolerance`	The pan-tilt is not corrected when the target center error is smaller than this pixel value
`kp_x`, `kd_x`	Pan-tilt horizontal control parameters
`kp_y`, `kd_y`	Pan-tilt tilt control parameters
`max_step_xy`	Maximum pan-tilt angle step per control cycle
`enable_body_turn_follow`	Whether the chassis is allowed to turn following the pan-tilt horizontal deviation
`body_kp`, `body_kd`	Chassis turning control parameters
`max_angular_z`	Maximum chassis angular velocity

4.3 Core Source Code Analysis

The main program file is:

```
yolo_onnx_follow_pkg/yolo_onnx_follow_node.py
```

Its core flow can be summarized as:

```
Load config -> initialize ROS topics -> load the ONNX model -> receive camera
images
-> YOLO inference -> NMS filtering -> assign tracking IDs -> select/re-acquire
the target
-> compute the pan-tilt angle -> compute the chassis angular velocity -> publish
control commands
```

4.3.1 Detection Data Structure

`Detection` stores the information of each detection box:

```python
@dataclass
class Detection:
    bbox: tuple[float, float, float, float]
    score: float
    class_id: int
    class_name: str
    track_id: int = -1
```

Where:

- `bbox`: detection box coordinates in the format `(x1, y1, x2, y2)`.
- `score`: confidence score.
- `class_id`: class index.
- `class_name`: class name.
- `track_id`: the temporary tracking ID assigned to the target internally by the program.

`cx`, `cy`, `width`, and `height` are property functions that quickly give the target center and box size; the subsequent pan-tilt control mainly uses `cx` and `cy`.

4.3.2 SimpleIouTracker Tracking ID Assignment

`SimpleIouTracker` is a lightweight local tracker that does not depend on any external tracking library. Its purpose is to keep the same object in consecutive frames assigned the same ID as much as possible.

The core logic is in `update()`:

```python
tracked = self.tracker.update(
    detections, image_width=frame.shape[1], image_height=frame.shape[0]
)
```

Matching order:

1. Compute IoU against same-class targets first; inherit the old ID when the IoU is greater than `track_iou_threshold`.

2. If IoU matching fails, use the center-point distance between same-class targets as a fallback match.
3. If it still does not match, assign a new `track_id`.

This reduces follow jitter when same-class objects move fast, when detection boxes suddenly grow or shrink, or when IDs briefly jump.

Related configuration:

```
track_iou_threshold: 0.2
track_center_threshold_ratio: 0.12
track_max_lost_frames: 30
```

A larger `track_center_threshold_ratio` makes it easier to join nearby same-class targets into the same ID; however, if several same-class targets are close together in the frame, a value that is too large may cause wrong matches.

4.3.3 Model Loading `_load_model()`

`_load_model()` checks the model path and loads the ONNX model:

```
model_path = str(self.demo_config.get("model_path", "") or "")
if not model_path or not os.path.exists(model_path):
    raise FileNotFoundError(...)
self.net = cv2.dnn.readNetFromONNX(model_path)
```

If `use_cuda` is `true`, the program tries to use the OpenCV DNN CUDA backend:

```
self.net.setPreferableBackend(cv2.dnn.DNN_BACKEND_CUDA)
self.net.setPreferableTarget(cv2.dnn.DNN_TARGET_CUDA_FP16)
```

If the current OpenCV has no CUDA support, keep the following in the configuration:

```
use_cuda: false
```

4.3.4 Image Callback `_image_callback()`

After the camera image enters the program, it is first converted to an OpenCV image with `cv_bridge`:

```
frame = self.bridge.imgmsg_to_cv2(msg, "bgr8")
```

The maximum inference rate is then limited by `max_inference_rate_hz` to avoid saturating the CPU by inferring on every frame:

```
if now - self.last_inference_time < 1.0 / max_rate:
    return
```

The actual recognition and tracking flow is:

```
detections = self._run_yolo(frame)
tracked = self.tracker.update(detections, image_width=frame.shape[1],
image_height=frame.shape[0])
self._update_selected_target(tracked, frame.shape[1], frame.shape[0])
```

In other words: YOLO detection first, then ID assignment, and finally updating the target to follow.

4.3.5 YOLO Inference `_run_yolo()` and Result Parsing `_decode_outputs()`

`_run_yolo()` first letterboxes the original image to the model input size:

```
letterboxed, scale, pad_x, pad_y = self._letterbox(frame, input_w, input_h)
```

It then builds the blob and runs the ONNX forward inference:

```
blob = cv2.dnn.blobFromImage(letterboxed, 1.0 / 255.0, (input_w, input_h),
swapRB=True, crop=False)
self.net.setInput(blob)
outputs = self.net.forward()
```

`_decode_outputs()` does four things:

1. Calls `_parse_yolo_row()` to parse each output row.
2. Filters low-confidence results with `conf_threshold`.
3. Removes duplicate boxes with `cv2.dnn.NMSBoxes()`.
4. Sorts by confidence and keeps only `max_detections` targets.

The current default configuration is:

```
conf_threshold: 0.2
max_detections: 3
```

So the window shows at most the 3 highest-confidence targets, reducing interference from cluttered detection boxes during selection and tracking.

4.3.6 Target Selection and Re-acquisition

There are two main entry points for target selection:

- `_on_mouse()`: after clicking a detection box in the window, records the target's `track_id` and class.
- `_select_detection()`: finds the target that should currently be followed from the detection results each frame.

After clicking a target, the program saves:

```
self.selected_track_id = int(detection.track_id)
self.selected_class_id = int(detection.class_id)
self.selected_last_bbox = detection.bbox
```

If the same `track_id` can still be found in the next frame, it keeps following directly. If the ID has jumped, it enters `_try_reacquire_selected_detection()` to find the target again by the same class and the previous frame's position.

The re-acquisition score considers three factors:

- IoU with the previous frame's detection box.
- Distance from the previous frame's target center.
- The current detection box's confidence.

Related configuration:

```
reacquire_enable: true
reacquire_iou_threshold: 0.05
reacquire_max_center_ratio: 0.45
reacquire_max_lost_frames: 30
reacquire_score_weight: 0.15
```

`_smooth_target_center()` smooths the target center point:

```
target_center_smoothing_alpha: 0.65
```

The smaller this parameter, the smoother the target center, but the slower the response; the closer to `1.0`, the faster the response, but the more visible the frame detection jitter.

4.3.7 OpenCV Display and Keys

`_show_frame()` draws the detection boxes, class names, IDs, and status information:

- Regular detection boxes are green.
- The currently followed target is yellow.
- The top shows the current frame count, detection count, image latency, and inference time.

`_handle_window_key()` handles keyboard input:

```
if key in (ord("r"), ord("R")):
    self._clear_selection()
elif key in (ord("q"), ord("Q"), 27):
    self.shutdown_requested = True
```

Pressing `r` clears the current target and publishes zero velocity; pressing `q` or `Esc` exits the program.

4.3.8 Pan-Tilt Following `_apply_ptz_follow()`

The input to pan-tilt following is the target center point:

```
err_x = float(state["cx"]) - half_w
err_y = float(state["cy"]) - half_h
```

If the error is smaller than `pixel_tolerance`, no correction is made to avoid small jitter:

```
if abs(err_x) <= cfg["pixel_tolerance"]:
    step_x = 0.0
```

Otherwise, the error is normalized and the angle step is computed with P+D control:

```
out_x = clamp(cfg["ptz_kp"] * norm_x + cfg["ptz_kd"] * deriv_x, -1.0, 1.0)
step_x = out_x * cfg["max_ptz_step"]
```

Finally, the pan-tilt angle is published through `PtzController.publish_servo_angles()`:

```
self.ptz.publish_servo_angles(
    servo1_angle=...,
    servo2_angle=...,
    repeat=False,
)
```

If the pan-tilt moves in the wrong direction, first change the following in the configuration:

```
sign_x: 1.0
sign_y: -1.0
```

4.3.9 Chassis Turning `_compute_body_turn()`

The chassis does not turn directly from the image pixel error; instead it turns according to the pan-tilt horizontal deviation. This way the pan-tilt chases the target first, and the chassis then slowly turns the car body back toward the target direction.

The judgment logic:

```
if abs(servo_err) >= cfg["servo1_engage_deg"]:
    self.body_correcting = True
elif abs(servo_err) <= cfg["body_tolerance_deg"]:
    self.body_correcting = False
```

That is:

- When the pan-tilt horizontal deviation exceeds `servo1_engage_deg`, the chassis starts turning.
- When the pan-tilt returns within the `body_tolerance_deg` range, the chassis stops turning.

Chassis angular velocity computation:

```
angular_z = cfg["body_dir"] * (
    cfg["body_kp"] * servo_err + cfg["body_kd"] * deriv
)
angular_z = clamp(angular_z, -cfg["max_angular_z"], cfg["max_angular_z"])
```

If the chassis turning direction is wrong, modify:

```
body_dir: -1.0
```

If the chassis turns too slowly, increase `body_kp` appropriately; if it turns too aggressively, reduce `body_kp` or `max_angular_z`.

4.3.10 PTZ Controller `ptz_controller.py`

`ptz_controller.py` encapsulates the pan-tilt angle publishing logic. The topic it publishes is:

```
/cmd_ptz_angle
```

The message type is:

```
geometry_msgs/msg/Vector3
```

Where:

- `Vector3.x`: horizontal servo angle.
- `Vector3.y`: tilt servo angle.
- `Vector3.z`: currently unused.

`publish_servo_angles()` clamps the angles with the servo limits before publishing:

```
self.servo1_angle = clamp(new_servo1_angle, self.servo1_min_angle,
self.servo1_max_angle)
self.servo2_angle = clamp(new_servo2_angle, self.servo2_min_angle,
self.servo2_max_angle)
```

On program exit, `recenter_after_stop()` returns the pan-tilt to its initial angle.